



Capstone Progress Report 1 Template

Report last updated: 2/20

Instructions: Each capstone team can use this template to capture and summarize progress. This can be shared with the sponsor and mentor. When submitting the report during the course, a PDF file is preferable.

Stakeholder Names and Roles

Stakeholder	Role
Ryan Healy	Team member
Louis Cocks	Team member
Katherine Kelleher	Team member
Anson Parker	Sponsor Role
Lane Rasberry	Sponsor Role
Rafael Alvarado	Sponsor Role

Project Title: *Wikipedia: Dispute/Arbitration Analysis and BPMN*

GitHub Repository: [rah-ds/Wikipedia_Dispute_Models: Capstone for Uva MSDS 2026](#)

Abstract

The objective of this project is to develop an automated framework for extracting formal ontologies and generating clear, executable process models from Wikipedia Arbitration Disputes. The research addresses the challenge of translating the raw Wikipedia disputes into machine-readable representations suitable for automated compliance checking, dispute workflow representation, and insight into the differences between stated and actual processes.

Outline of the Project

This project will demonstrate the use of Ontology Extraction and Business Process Modeling & Notation (BPMN) within a scoped and well-understood domain. The objective is to extract processes for Wikipedia arbitration by gathering documents, extracting and mining data, and using subject matter expertise to verify processes.

In practice, this project will provide Wikipedia users with (1) clarity on the various avenues in which disputes are often handled and (2) demonstrate a proof-of-concept analytical solution that can be

generalized and used to extract process information. The models produced for this project will help distill the complicated nuances of varied Wikipedia dispute processes, typically only known by Wikipedia experts, by visually representing processes as sequences of formal or informal events. The proof-of-concept solution and methodology for process extraction from extensive text data will provide value by demonstrating how a technical solution can collect and analyze information at scale to provide insight into processes that would otherwise be manually intensive or impossible to interpret.

The stakeholders for this project include Anson Parker (Lexipedia Project Lead) and Lane Rasberry (Wikipedia Dispute Subject-Matter-Expert). Anson Parker is looking to develop an analytical solution that demonstrates process extraction. Lane Rasberry has knowledge about the Wikipedia dispute process. The analysis for this project is conducted with the following assumptions:

- Every arbitration, no matter how big or small, inherently has some level of business process behind its arbitration.
- Arbitration is not solely comprised of argumentative statements.
- Wikipedia has dispute data that is accessible via API during the timeline of the project.
- Any information within Wikipedia Arbitration/Dispute pages is available and may be included in scope for process extraction (excluding non-English examples to help maintain consistency).
- Statistical insights related to disputes held within Wikipedia from any members/contributors, including at least two individuals, will also be included within the project's analysis.

To maintain scope and scalability, no personal individual information about users within a dispute is considered in scope for this project.

Success Criteria

SC1	<i>Automate process for pulling and cleaning data related to Wikipedia Dispute Process</i>
SC2	<i>Generate exploratory data analysis and statistical insights for corpus of text</i>
SC3	<i>Extract and visualize key dispute processes for documented Wikipedia disputes</i>
SC4	<i>Display data overview and select BPMN examples in an interactive dashboard</i>

Data Assumptions and Limitations

Due to the nature of Wikipedia dispute and arbitration data, the following limitations will affect the results of this project:

- Running BPMN and visualizing at scale may result in process abstraction.
- Cleaning diverse unstructured data formats will require subject matter expertise and is difficult to fully automate.
- Available data represents Wikipedia dispute processes from over 20 years of data, likely reflecting an evolving process over time

Given the limitations listed above and scope/nature of data extracted a few data assumptions are made for the purposes of this project's analysis:

- The arbitration process lineage extraction cannot account for dead links or a change in the host of a webpage

- This project cannot validate with perfect accuracy whether the data represents the true events and full scope of real-life, Wikipedia arbitration processes and disputes.

For the purposes of compute power and memory, University of Virginia's Rivanna (high-performance computing (HPC) cluster) will be utilized for pulling the entire data set with a slurm-script gathering all revisions of Requests for Comment (RfC), Edit Wars, Dispute Resolution Noticeboard (DRN) cases, and Arbitration Cases.

Summary of Data Processing, Data Aggregation

The raw data used for analysis in the Lexipedia Capstone Project is open source, unstructured information from Wikipedia. The data is pulled from Wikipedia pages via available Wikimedia APIs, and contains unstructured data related to informal and formal stages of the Wikipedia dispute resolution process. In particular, the automated data script pulls data from key stages available for the dispute resolution process, that reference other stages of the dispute resolution process, such as the Request for Arbitration, DRN pages, RfCs, Revisions, and Edit Wars.

Experiments are underway to pull all text lineage for an arbitration case via depth first search so that we can compare everything that goes into a case and the whole history of all actors involved.

These unstructured data sources contain the target of this project, which is a process, comprised of dispute stages and outcomes, in a text format. In particular, the data sources contain the following relevant variables:

- Case Opening/Closing date
- Confirmation that all parties are aware of the request
- Other steps in dispute resolution have been tried
- Requests for comment
- Statements
- Preliminary decisions
- Consequences

Initial data processing involves cleaning unstructured text into semi-structured format, aligning to the general, commonly used Wikipedia dispute templates. Additional data processing techniques will be evaluated to determine feasibility and utility for BPMN modeling, such as part-of-speech tagging, named entity recognition, and sentence segmentation.

Data Visualizations

The following images are snapshots taken from data extraction, summary statistics of the data, process flow mapping and categorization, and BPMN extraction scripts.

Figure 1: Unstructured Arbitration Data (JSON)

```

1  {
2    "title": "Wikipedia:Requests for arbitration/R11-",
3    "url": "https://en.wikipedia.org/wiki/Wikipedia:Requests_for_arbitration/R11-",
4    "last_edit": "2024-06-02T01:29:00Z",
5    "revisions": [
6      {
7        "revId": 1226822113,
8        "parentId": 1226788201,
9        "timestamp": "2024-06-02T01:29:00Z",
10       "user": "Steel1943",
11       "comment": "Steel1943 moved page [[Wikipedia:Arbitration/Requests/Case/R11-]] to [[Wikipedia:Requests for arbitration/R11-]] over a redirect without leaving a redirect: Undoing my recent moves to match parent page per recent developments at [[W:ABR]], not leaving redirects behind since the title did not exist before",
12       "size": 29267
13     },
14     {
15       "revId": 1226788201,
16       "parentId": 1142571915,
17       "timestamp": "2024-06-03T20:41:40Z",
18       "user": "Steel1943",
19       "comment": "Steel1943 moved page [[Wikipedia:Requests for arbitration/R11-]] to [[Wikipedia:Arbitration/Requests/Case/R11-]]: Match current name of parent page",
20       "size": 29267
21     },
22     {
23       "revId": 1142571915,
24       "parentId": 1039412991,
25       "timestamp": "2023-03-03T04:55:27Z",
26       "user": "MalnadachBot",
27       "comment": "Fixed [[WP:LINT|lint errors]]. ([[User:MalnadachBot/Task 12|Task 12]])",
28       "size": 29267
29     },
30     {
31       "revId": 1039412991,
32       "parentId": 910550411,
33       "timestamp": "2021-08-18T15:22:31Z",
34       "user": "MalnadachBot",
35       "comment": "Fixed [[WP:LINT|lint errors]] in signatures. ([[User:MalnadachBot/Task 2|Task 2]])",
36       "size": 29230
37     },
38   ]
39 }

```

Figure 2: Unstructured Requests for Comment Data (JSON)

[illegible]

Figure 3: Overview of Dispute-Related Records Captured 2/16/26-2/20/26

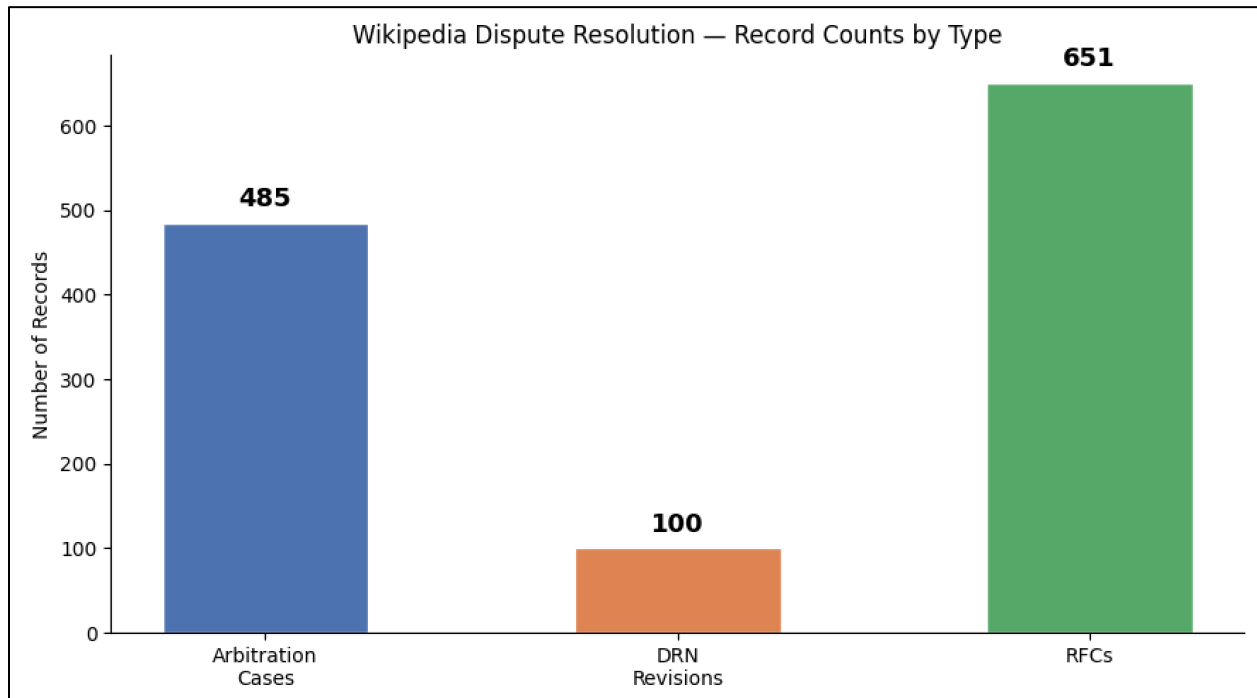


Figure 4: Exploratory Escalation Funnel of Wikipedia Arbitration

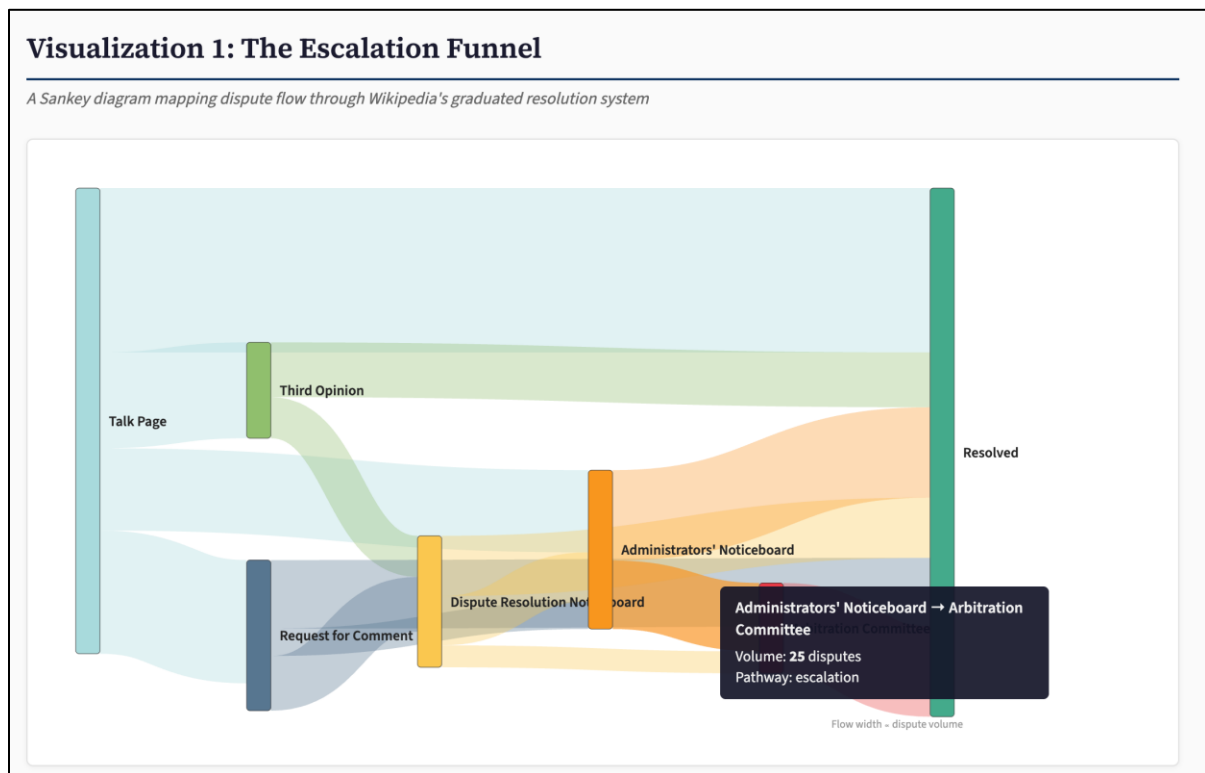


Figure 5: Categorization of Dispute Topics Based on Textual Analysis

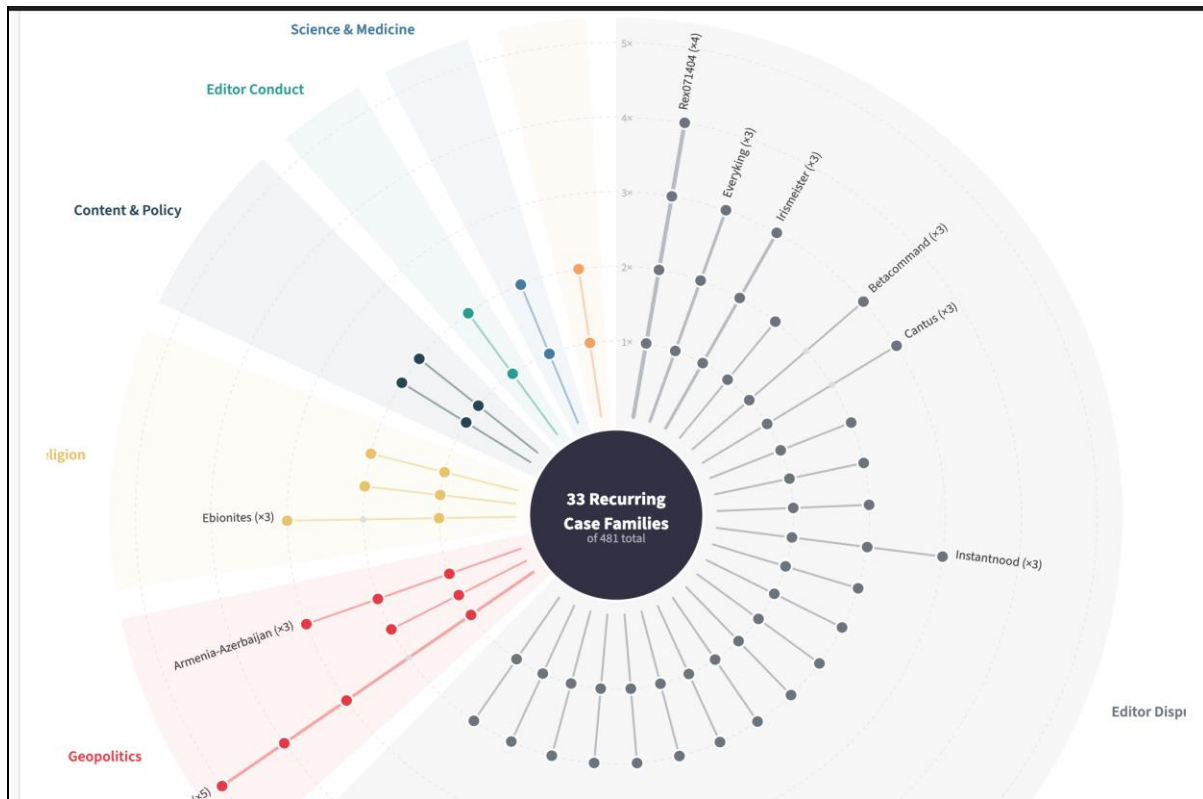


Figure 6: Dispute Resolution Noticeboard (DRN) Aggregate Process Flow

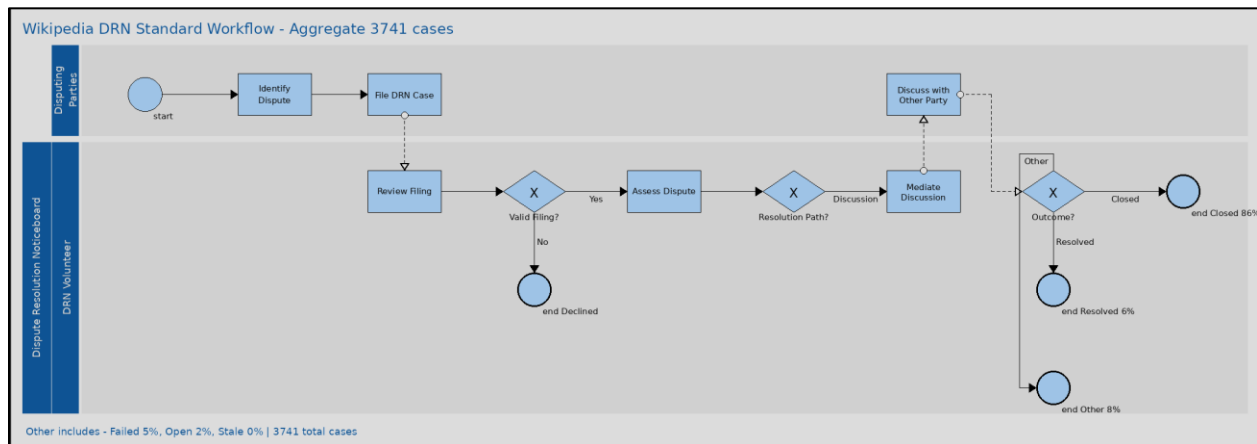


Figure 7: Dispute Resolution Noticeboard (DRN) Example Process Flow – Power Electronics

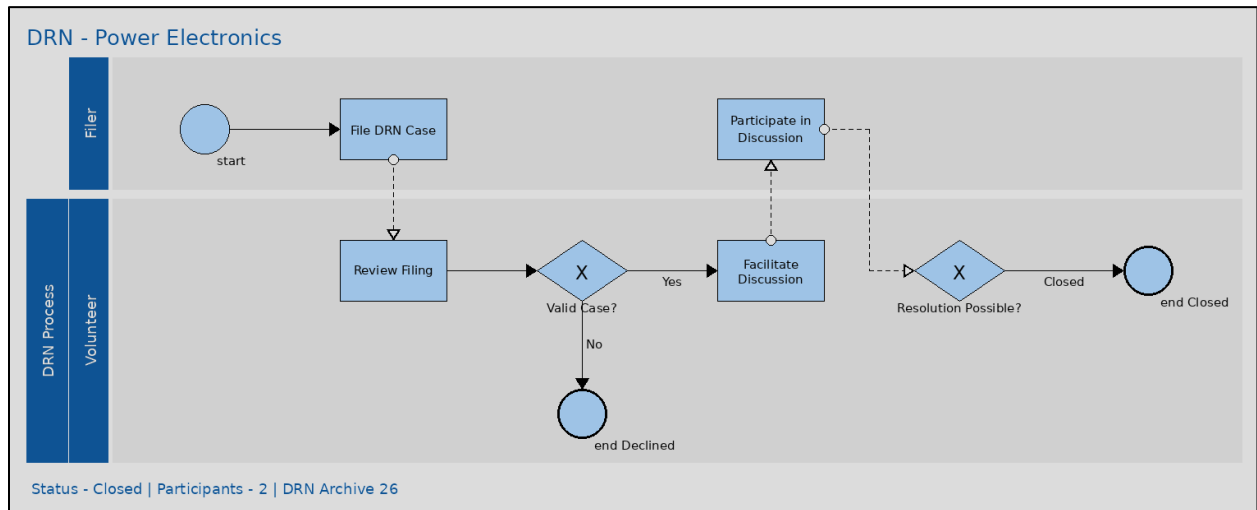


Figure 8: Request for Comment (RfC) Aggregate Process Flow

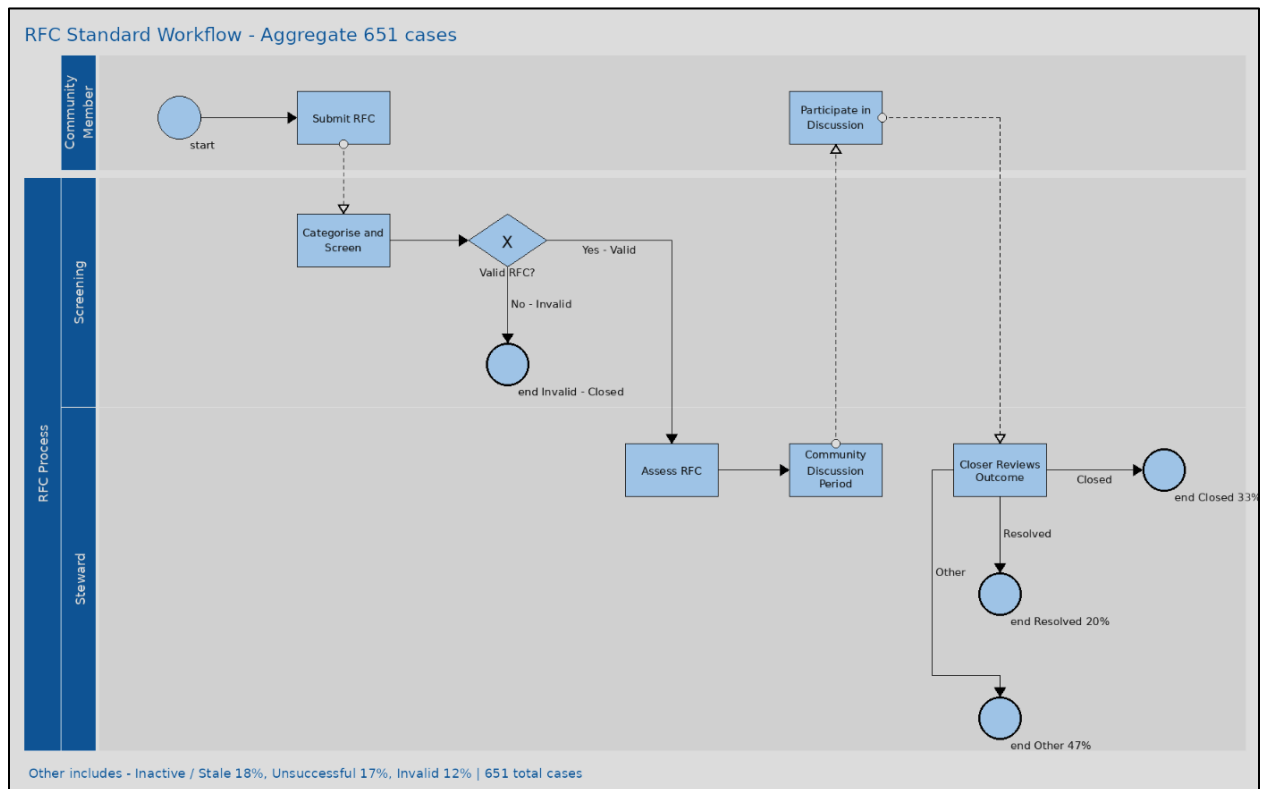
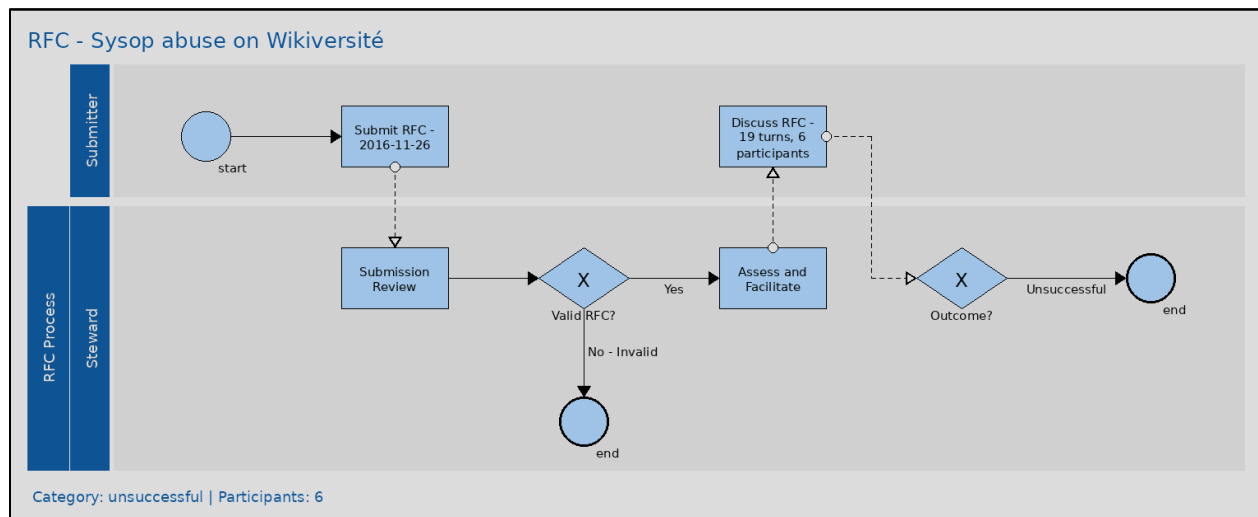


Figure 9: Request for Comment (RfC) Example Process Flow – Sysop Abuse on Wikiversité



Summary of Modeling and Analysis

To accomplish process extraction from natural language documents, several models are being considered. The first is natural language processing (NLP) models such as Text-to-BPMN, which supports the extraction of process extraction with coded logic and procedural rules based on textual references to known events in the arbitration process. An evaluation will be conducted with prompts and several common large-language models to assess their utility for process extraction with a few arbitration cases.

A few cases will be validated for references to arbitration stages, with a focus of minimizing false positives and precision. This evaluation will focus on these metrics in order to prioritize accurately modeling events that appear in text.

Future Work Plan

Task Description	Date
Slurm script for pulling all the data into UVA compute over a few days.	April 1
(Preloaded) Develop fully automated script to from pulling a set of Wikipedia data	April 10
Build BPMN models from the data examples pulled	April 10
Finalize data processing and tagging for BPMN refinement	April 10
Build interactive visuals in a dashboard (data overview and BPMN)	April 10
Create instructional README for repeating Wikipedia data pull and analysis	April 28
Final Presentation Materials (slide deck)	April 30
(Reach) Attempt LLM prompting for model building from extracted data.	N/A

Potential Concerns [C] and Blockers [B]

The unstructured nature of Wikipedia dispute process data is the largest challenge for this project. The dispute cases can have varied structures, so the final solution will prioritize capturing a high-level overview of processes at scale. Focus on detailed events within processes will be included in the individual analysis for select cases.

Identifier	Description
Dashboard	<i>Looking for a dashboard hosting solution like Heroku</i>
Rivanna	<i>Maintaining the Rivanna cluster that will host our data.</i>